Emittance Measurements at ALBA Linac:

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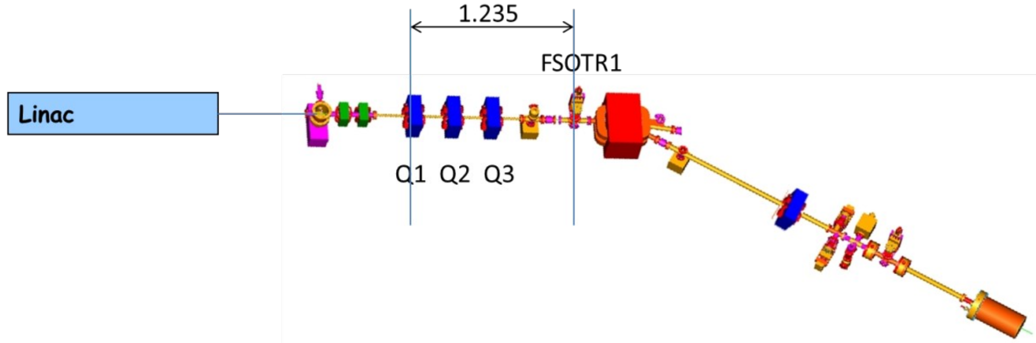
Experimental Set-up:



Sketch of the Linac, the transfer line and the Diagnostics line.

Note:

- Beam energy: E~80MeV – to be crosschecked in the Control Room
- Intensity of dipole power supply: I_{bend} ~ 110A
- Dipole calibration:
  - $E = A'*I_{bend} + A'_0$
  - A'=0.7058 MeV/A ; A'_0= 0.2448 MeV
- Intensity of quad power supply: I_{quad} ~[-5, +5]A
- Distance Q1 → FSOTR: d=1.235m
- Quadrupole length: L_Q = 0.126m
- Quadrupole calibration:
  - Quadrupole calibration factor, F=387.65 MeV/A/m^2
  - Quadrupole strength [1/m^2]: $K_1 = I_{quad}*F/E$
  - Quadrupole integrated strength [1/m]: $k=K_1L_Q$
  - Quadrupole strength square root [1/m]: $s_k = \mathrm{sign}(K_1)*\mathrm{sqrt}(|K_1|)$
  - Quadrupole focal strength [1/m]: $Q_{21} = s_k*\sin(|s_k L_Q|)$

Beam Characteristics Multi-Bunch-Mode

1. Look at the Linac Beam Charge Monitor (BCM) and Fast Current Trasnformer, and determine the longitudinal characteristics of the beam.

a) What is the beam charge?
b) How long is the bunch train?
c) How many bunches are in the train?

2. After the beam passes the accelerating structures, the beam energy is around 80MeV.
Write down the dipole magnet intensity, and calculate the beam energy using the calibration given in pg.1. Compare it with the results from the specific GUI.

Horizontal Emittance Measurement Multi-Bunch-Mode

3. Scan the intensity of Q1, I_{quad}, until you find a beam waist (minimum beam size) in the horizontal plane, and fill the table below:

	I_{quad}, A	Q_{21}, m^{-1}	$K_1 \cdot L_Q$, m^{-1}	σ, mm	σ^2, mm^2	% diff thin lens*
Waist + 2*ΔI						
Waist + ΔI						
Waist						
Waist - ΔI						
Waist - 2*ΔI						

* % diff thin lens = $(K_1L_Q-Q_{21})/Q_{21}*100$

4. Plot $Q_{21} = s_k*\sin(s_k L_Q)$, ($s_k = \text{sign}(K_1)*\text{sqrt}(|K_1|)$) vs σ^2, and fit the data to the following Eq. to find A, B, C:

$$\sigma^2 = A\,(k - B)^2 + C$$

a. From the parameters A, B, C, obtain the so-called geometric emittance using
$\varepsilon = \text{sqrt}(A*C)/d^2$

b. What is the normalized emittance?

5. Compare the quadrupole focal strength values Q_{21} with the integrated quadrupole strength $k=K_1L_Q$. Is the thin lens a good approximation?

Vertical Emittance Measurement Muclti-Bunch-Mode

Repeat steps [3-5] for the Vertical plane.

In this case, the table is:

	I_{quad}, A	Q_{21}, m^{-1}	$K_1 \cdot L_Q$, m^{-1}	σ, mm	σ^2, mm^2	% diff thin lens
Waist + 2*ΔI						
Waist + ΔI						
Waist						
Waist - ΔI						
Waist - 2*ΔI						

Beam Characteristics Single-Bunch-Mode

1. After switching to single bunch operation, look at the Linac Beam Charge Monitor (BCM) and Fast Current Trasnformer, and determine the longitudinal characteristics of the beam.

d) What is the beam charge?
e) How long is the bunch train?
f) How many bunches are in the train?

2. After the beam passes the accelerating structures, the beam energy is around 80MeV.
Write down the dipole magnet intensity, and calculate the beam energy using the calibration given in pg.1. Compare it with the results from the specific GUI.

Horizontal Emittance Measurement Single-Bunch-Mode

Repeat steps [3-5] for the Horizontal plane.

	I_{quad}, A	Q_{21}, m^{-1}	$K_1 \cdot L_Q$, m^{-1}	σ, mm	σ^2, mm^2	% diff thin lens
Waist + 2*ΔI						
Waist + ΔI						
Waist						
Waist - ΔI						
Waist - 2*ΔI						

Vertical Emittance Measurement Single-Bunch-Mode

Repeat steps [3-5] for the Vertical plane .

	I_{quad}, A	Q_{21}, m^{-1}	$K_1 \cdot L_Q$, m^{-1}	σ, mm	σ^2, mm^2	% diff thin lens
Waist + 2*ΔI						
Waist + ΔI						
Waist						
Waist - ΔI						
Waist - 2*ΔI						

Comparison about emittance measurements

Compute the normalized emittance in both cases and explain the results.